

MedNexus Master Plan

Architecture & Implementation Blueprint

1. The Core Architecture: Tags, Not Folders

Your database does not use physical "folders." It is a single, flat table of questions organized entirely by tags and statuses.

- **The Tag System:** Every question is assigned a `module_id` (e.g., Cardiology) and an optional `discipline_id` (e.g., Internal Medicine).
- **The Status Cascade:** You have absolute control over visibility.
 - **Module Level (Live / Offline):** If an entire module is set to Offline, it disappears from the student's app entirely. This lets you build massive question banks in the background.
 - **Question Level (Live / Draft / Offline):** When a module is Live, students only see the specific questions inside it that are *also* marked Live.

2. The Student UI: "Study Modules" & The View Toggle

We give the student instant control over their library without duplicating any data.

- **The Toggle Placement:** Right below the top header, you place a  toggle:

[ View by Body System (Module)] | [ View by Specialty (Discipline)]

- **How it works:**
 - When **Module** is active, the grid renders cards like Cardiology and Dermatology. Clicking one reveals the sub-disciplines inside it.
 - When **Discipline** is active, the React frontend instantly re-sorts those exact same questions. The cards display specialties (Internal Medicine, Surgery). Clicking one reveals the body systems attached to that specialty.

3. The Admin UI: The Question Editor Dashboard

Your admin side is built for speed and bulk management. It abandons the "card" look for a high-density **Data Table**.

- **The Control Bar:** At the top, you have a Search bar, followed by strict dropdown filters:
[Module ▼], [Discipline ▼], and [Status ▼].
- **Primary Actions:** [ Import PDF], [ Export JSON], and [+ Add Question].

- **Bulk Actions & JSON:** The data table features checkboxes next to every row.
 - Selecting multiple rows summons a floating action bar: [Set Offline], [Change Module], [Delete].
 - Clicking [📄 Export JSON] downloads *only* the filtered, selected rows. The tags export with it for easy re-import.

4. Adding vs. Editing Questions (Modals & Drawers)

A. Adding a New Question (The Center Pop-up Modal)

- Clicking [+ Add Question] opens a focused, central pop-up.
- **Inline Tag Creation:** The Module and Discipline fields are searchable comboboxes. If you type "Endocrinology" and it doesn't exist, a [+ Create "Endocrinology"] button appears inside the dropdown to instantly save and select the new tag.

B. Editing Existing Questions (The Continuous Side Drawer)

- Clicking a question row in the Admin Table slides open a wide **Side Drawer** from the right edge of the screen, giving room to read full clinical vignettes.
- **The Flow:** The top features [← Previous] | [Next →] buttons. You can fix a typo, hit "Save," and click "Next" to instantly load the next question without returning to the main table.

5. The PDF Import & Staging Workflow

To import hundreds of questions flawlessly, rely on an AI parser combined with a staging area.

The Golden PDF Template

Format your offline documents with strict bracket tags so the AI never guesses:

```
[MODULE: Cardiology]
[DISCIPLINE: Internal Medicine]

1. A 55-year-old man from Cape Coast presents with...
A. Option 1
B. Option 2
ANS: A
EXP: Rationale goes here...
```

The Staging Area

When uploaded, these parsed questions enter your Admin Table marked as **Draft**. You review the list, quickly fix any tags if needed, select all, and flip their status to **Live**.